Yuxin Jin

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Education

University of Wisconsin-Madison 2025/09 - 2027/05 (Expected)
Bachelor of Arts in Computer Science Madison, WI, USA
- Dean's List **(GPA: 4.0/4.0)**

Xi'an Jiaotong-Liverpool University 2023/09 - 2025/06 (Transferred Out)
B.Eng in Data Science and Big Data Technology Suzhou, Jiangsu, China
- University Academic Achievement Award, 2025. **(10%)**
- University Academic Excellence Award, 2024. **(5%)**

Research Interests

I am broadly interested in **learning from imperfect supervision**, including weak-to-strong generalization, data-centric methods for training foundation models, and LLM post-training with imperfect reward signals. I am also interested in evaluating the reliability and robustness of foundation models.

Research Experience

Sprocket Lab, University of Wisconsin-Madison 2026/05 - present
Undergraduate Researcher, advised by Prof. Frederic Sala; mentored by John Cooper Madison, WI, USA
- Co-developed RADS, a representation-alignment data selection method for weak-to-strong (W2S) generalization that prioritizes points with features the strong student encodes but the weak teacher misses; across 9 datasets, RADS achieved a mean performance gap recovered (PGR) of 0.62, exceeding the strongest non-RADS baseline by 0.1
- Wrote and maintained the training and evaluation code (weak-label generation, LoRA fine-tuning, excess-loss and random-selection baselines), ran experiments on 23 datasets, and built a training-free CKA probe that ranks datasets by how much the selected half outperforms the rest (Spearman 0.84); co-first author of the paper under review at ICLR 2027

Intelligent Ultrasound Research Center, East China Normal University 2025/09 - 2026/04
Undergraduate Researcher, advised by Prof. Jiangang Chen; mentored by Renxing Li Remote
- Led a review of AI-assisted musculoskeletal ultrasound (MSUS) for aromatase inhibitor–associated arthralgia (AIA): designed the systematic search across 3 databases and a trial registry, screened 816 records down to 129 included studies, and wrote the AI-focused sections on ML/DL methods for ultrasound automation
- Headed a 5-member student research team and coordinated with the hospital: held bi-weekly meetings with professors and clinicians to translate clinical pain points into the review scope and key questions, and drove the team to finalize the outline and complete the manuscript workflow

School of AI and Advanced Computing, Xi'an Jiaotong-Liverpool University 2024/12 - 2025/06
Undergraduate Researcher, advised by Prof. Jionglong Su; mentored by Haochen Xue Suzhou, Jiangsu, China
- Contributed to the construction and curation of a multimodal real-world dialogue dataset (5,000+ conversations) with six types of manually labeled questions, and supported benchmarking of 20 MLLMs, observing that models likely answered correctly when the needed fact was stated but otherwise could fail to decline questions about unmentioned things
- Conducted literature search and summarization, participated in weekly research meetings, drafted core manuscript sections, produced figures, and supported multiple rounds of revisions for a review of adversarial attacks and defenses for LLMs

Publications

- **Aligning Prediction Residuals: Representation-Aware Data Selection for Weak-to-Strong Generalization**. Cooper, J.[*], Huang, T.-H.[*], **Jin, Y.**[*], Bastian, N. D., Vinayak, R. K., Sala, F. ([*] equal contribution). Under review at *International Conference on Learning Representations* (**ICLR 2027**).
- **Toward Artificial Intelligence–Assisted Musculoskeletal Ultrasound for Aromatase Inhibitor–Associated Arthralgia: Evidence, Gaps, and a Research Agenda**. **Jin, Y.**[*], Zheng, S.[*], Wang, Y., Chang, J., Dong, L. ([*] equal contribution). Manuscript in preparation.

- **Adversarial Attacks and Defenses on Large Language Models: A Systematic Review**. Li, R., Liu, J., Zhen, J., Bao, Y., Qiang, Y., Xu, Y., Gu, Y., **Jin, Y.** (8th/13), Yu, D., He, Y., Zhou, M., Stefanidis, A., Su, J. (2025). *8th International Conference on Big Data and Artificial Intelligence* (**BDAI 2025**). [Paper]

Project Experience

Minimind: End-to-End Small LLM Training & Alignment 2026/02

- Reproduced and extended a full LLM pipeline (pretraining -> SFT -> LoRA -> DPO -> PPO) for a 26M-parameter bilingual language model on a single 12 GB GPU (RTX 4070) with PyTorch; fine-tuned on ~1.2 GB of SFT dialogues, trained DPO on 17k preference pairs, and ran PPO through 5 iterative cycles of policy optimization with an external 1.8B reward model
- Designed a controlled parallel-branch evaluation: compared 3 post-training methods (LoRA, DPO, PPO) from the same SFT checkpoint on a fixed 8-prompt suite; and documented findings in 5 structured stage reports and 1 mind map

Course Project: Relational Database Design & SQL Analytics 2025/04 - 2025/06

- Designed a PostgreSQL database for a movie analytics platform (requirements -> ERD -> 3NF -> physical deployment), integrating ~10k records across films, people, roles, and reviews
- Enforced integrity via PK/FK, composite keys, UNIQUE and CHECK constraints; implemented audit logging and versioned review snapshots with soft deletion for traceability under privacy considerations
- Imported and standardized multi-table CSV datasets, resolving ingestion issues (date parsing failures, floating-point precision drift, type inconsistencies) and implementing analytical queries for ranking, filtering, and aggregation

Course Project: Deep Learning from Scratch 2025/02 - 2025/04

- Implemented five end-to-end labs from raw data to evaluation: PyTorch MLP on MNIST digits, scikit-learn Logistic Regression on Iris, scikit-learn MLPRegressor on housing prices, TensorFlow RNN/LSTM for stock forecasting, and Keras CNN for Fashion-MNIST image classification
- Collected and annotated a 10-class web-crawled image dataset (1800+/200/220 train/val/test), trained a YOLO-based model, and evaluated on the validation set (200 images, 500+ instances) with mAP@0.5=0.34 and mAP@0.5:0.95=0.23, using confusion matrix and prediction visualizations to analyze error cases

Technical Skills

- **Languages:** Python, C/C++, Java, SQL, Bash
- **Frameworks and Tools:** PyTorch, Transformers, PEFT (LoRA), scikit-learn, TensorFlow, CUDA, Linux, Git, LaTeX

Last Updated on October 28, 2026